

Why Kapustka's degree-20 contraction does not embed in $\mathbb{P}^7$, and what that does and does not say about $\mathrm{SR}(\mathcal{M})$

Note for Daniel (run runs/kapustka-degree20-tangent-space-note-2026-09-18)

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Scope. One limited question: the contracted threefold $\bar{Y}$ in G. Kapustka's construction of a degree-20 Calabi–Yau threefold (arXiv:1010.3895, Table 1, No. 8) has a singular point with a nine-dimensional Zariski tangent space. Why is that true, why does it stop $\bar{Y}$ from lying in $\mathbb{P}^7$, and what does it mean for comparing Kapustka's smooth fibres with a smoothing of the Stanley–Reisner scheme $\mathrm{SR}(\mathcal{M})$? The smoothing of $\mathrm{SR}(\mathcal{M})$ is under review and is *not* assumed here. Every computation quoted below was rerun exactly over $\mathbb{Q}$ unless it is explicitly marked as a finite-field check; the logs are in `logs/`, the sources and quotations in `SOURCES.md`.

1 The objects

The sphere side

$\mathcal{M}$ is the Grünbaum–Sreedharan eight-vertex triangulation of S^3 . In the labelling of the repository its Stanley–Reisner ideal in $S = k[a, \dots, h]$ is generated by the sixteen cubic monomials

$$\begin{aligned} abf, abg, abh, acg, ach, adh, bdf, bdg, \\ beg, cde, ceg, ceh, cfh, def, dfh, efg, \end{aligned}$$

and $\mathrm{SR}(\mathcal{M}) = \mathrm{Proj} S/I_{\mathcal{M}} \subset \mathbb{P}^7$ is the union of the twenty coordinate $\mathbb{P}^3$'s indexed by the facets. Exactly over $\mathbb{Q}$ (`logs/c01`): $S/I_{\mathcal{M}}$ has Hilbert function 1, 8, 36, 104, 232, 440, Hilbert polynomial $P(n) = \frac{10}{3}n^3 + \frac{14}{3}n$, degree 20, no quadrics, and minimal free resolution with Betti numbers 1, 16, 30, 16, 1 in degrees 0, 3, 4, 5, 8.

Definition 1.1. A closed subscheme $Z \subset \mathbb{P}^N$ is *arithmetically Gorenstein* (aG) if its homogeneous coordinate ring S/I_Z is a Gorenstein ring: it is Cohen–Macaulay (its minimal free resolution over S has length $\mathrm{codim} Z$) and the last module of that resolution has rank one. This is a property of the *embedding*. It is stronger than Z being *Gorenstein*, which is the local property that every local ring of Z is Gorenstein (smooth points, hypersurface singularities and quotients of these by groups acting trivially on the canonical form are Gorenstein).

So $\mathrm{SR}(\mathcal{M})$ is aG in $\mathbb{P}^7$ with h -vector (1, 4, 10, 4, 1), and the last resolution degree 8 says $\omega_{\mathrm{SR}(\mathcal{M})} \cong \mathcal{O}$. At a vertex, say $a = 1$, all other seven coordinates are local equations of degree ≥ 2 ($\mathcal{M}$ is neighbourly), so the Zariski tangent space there is all of $\mathbb{A}^7$ (`logs/c01`): $\mathrm{SR}(\mathcal{M})$ is as singular as a subscheme of $\mathbb{P}^7$ can be at those points, but it lives in $\mathbb{P}^7$ by construction.

Kapustka's construction

A *del Pezzo surface* is a smooth projective surface with $-K$ ample; its degree is K^2 . We need only $D = \mathbb{P}^1 \times \mathbb{P}^1$, with $-K_D = \mathcal{O}(2, 2)$, $K_D^2 = 8$. The nine bidegree-(2, 2) monomials embed D in $\mathbb{P}^8$ as a surface D_8 of degree 8, cut out by twenty quadrics (`logs/c02`); this is the *anticanonical embedding*. Kapustka projects D_8 from a general line of $\mathbb{P}^8$ to $S \subset \mathbb{P}^6$. The projection is an isomorphism $D \cong S$ (a general line misses the five-dimensional secant variety), but S is *not linearly normal*: the seven coordinates of $\mathbb{P}^6$ restrict to a 7-dimensional subspace V of the 9-dimensional $H^0(S, \mathcal{O}_S(1)) = H^0(D, \mathcal{O}(2, 2))$. Write $h^1(I_S(1)) = 9 - 7 = 2$ for the deficit. Over $\mathbb{Q}$, for a random integer centre (`logs/c03`): I_S has three quadrics and fourteen cubics, the Betti table Kapustka prints in the proof of his Theorem 5.1, and

$$h^1(I_S(k)) = 0 \quad \text{for } k \geq 2,$$

so $k[S]_n = H^0(D, \mathcal{O}(2n, 2n))$ for $n \geq 2$. Since $h^1(I_S(k))$ is upper semicontinuous in the centre, these values hold for the general line.

Then: $X' \subset \mathbb{P}^6$ is a general complete intersection of two quadrics and a cubic containing S , a Calabi–Yau threefold of degree 12 with 44 nodes on S (Kapustka, Table 1; the count was re-derived and finite-field checked in the 2026-09-11 run). Blowing up S gives a small resolution, and flopping its 44 exceptional curves gives a smooth Calabi–Yau threefold X with $\mathrm{Pic} X = \mathbb{Z}H^* \oplus \mathbb{Z}D'$, where H^* pulls back $\mathcal{O}_{X'}(1)$ and $D' \cong D$ is the strict transform of S . Because $K_X = 0$, adjunction gives

$$\mathcal{O}_{D'}(D') = K_{D'}, \quad \mathcal{O}_{D'}(H^*) = \mathcal{O}_D(2, 2) = -K_{D'},$$

so the divisor $G := H^* + D'$ restricts trivially to D' . Sections: from $0 \rightarrow \mathcal{O}_X(H^*) \rightarrow \mathcal{O}_X(G) \rightarrow \mathcal{O}_{D'} \rightarrow 0$ and $H^1(\mathcal{O}_X(H^*)) = 0$ (Kawamata–Viehweg, H^* nef and big, $K_X = 0$),

$$H^0(X, G) = s \cdot H^0(X, H^*) \oplus \langle w \rangle, \quad \dim = 7 + 1 = 8,$$

where s is the section defining D' , the seven sections $s x_i$ vanish on D' , and w does not. This is the $h^0(H) = 8$ of Table 1.

Three different objects

- (a) The *normal contraction* $\varphi : X \rightarrow \bar{Y}$: the morphism with connected fibres that contracts exactly D' to a point P (Kapustka takes $\varphi = \varphi_{|2G|}$ in Theorem 3.4 and says in the proof of Theorem 5.1 that the degree-20 case “concludes as in the proof of Theorem 3.4”). It is a *primitive contraction of type II* in the sense of Wilson and Gross: birational onto a normal variety, not factorisable, with exceptional locus a divisor mapping to a point. There is an ample Cartier divisor T on $\bar{Y}$ with $\varphi^*T = G$, so $H^0(\bar{Y}, T) = H^0(X, G)$ has dimension 8 (Kapustka, proof of Theorem 3.4: “the image T of G on Y is an ample divisor ($2T$ is very ample)”).
- (b) The *image under the eight sections* $Y := \varphi_{|T|}(\bar{Y}) \subset \mathbb{P}^7$.
- (c) The *smooth fibres* Y_t of a smoothing $\mathcal{Y} \rightarrow \Delta$ of $\bar{Y}$, which exists by Gross, *Deforming Calabi–Yau threefolds*, Theorem 5.8: a primitive type II contraction is smoothable unless the exceptional divisor is $\mathbb{P}^2$ or $\mathbb{F}_1$. Table 1 records $H^3 = 20$, $h^0(H) = 8$, $\chi = -60$, and $h^{1,1} = 1$, $h^{1,2} = 31$.

Kapustka’s Remark 3.3, for the analogous degree-15 row, says exactly that (b) is not normal at P and that one must pass to a multiple of G to obtain (a). The point of this note is what happens at P .

2 The tangent space at P is nine-dimensional

Recall (HP, algebraic geometry, “Zariski tangent space”) that for a point P with local ring $(\mathcal{O}_P, \mathfrak{m})$ the Zariski tangent space is $T_P = (\mathfrak{m}/\mathfrak{m}^2)^\vee$. Its dimension is the *embedding dimension*: if $P \in Z$ and Z is a closed subscheme of a smooth N -fold near P , then the cotangent space of the ambient space surjects onto $\mathfrak{m}/\mathfrak{m}^2$, so $\dim T_P Z \leq N$.

Theorem 2.1. $\dim T_P \bar{Y} = 9$ and $\text{mult}_P \bar{Y} = 8$. More precisely, the germ $(\bar{Y}, P)$ is analytically the affine cone over $D_8 \subset \mathbb{P}^8$, that is,

$$(\bar{Y}, P) \cong (\{xy - zw = 0\} \subset \mathbb{A}^4)/\{\pm 1\},$$

the three-dimensional ordinary double point divided by the sign involution.

Step 1: the lower bound $\dim T_P \bar{Y} \geq 9$ by hand. Let $\hat{\mathcal{O}}$ be the completion of $\mathcal{O}_{\bar{Y}, P}$. The theorem on formal functions (Hartshorne III.11.1), applied to the proper morphism φ whose fibre over P is D' , gives

$$\hat{\mathcal{O}} = \varprojlim_n H^0(X, \mathcal{O}_X/\mathcal{O}_X(-nD')).$$

(The theorem uses the powers of $\mathfrak{m}\mathcal{O}_X$; these are cofinal with the powers of the ideal sheaf $\mathcal{O}_X(-D')$ of the reduced fibre, so the limits agree.) For $n \geq 1$ the quotient $\mathcal{O}_X(-nD')/\mathcal{O}_X(-(n+1)D')$ is the line bundle $\mathcal{O}_{D'}(-nD') = \mathcal{O}_D(2n, 2n)$, whose H^1 vanishes; so all transition maps in the inverse system are surjective and so is $\hat{\mathcal{O}} \rightarrow H^0(\mathcal{O}_X/\mathcal{O}_X(-2D'))$. Put $F_n := \ker(\hat{\mathcal{O}} \rightarrow H^0(\mathcal{O}_X/\mathcal{O}_X(-nD')))$, the functions vanishing to order $\geq n$ along D' . Then $F_1 = \hat{\mathfrak{m}}$ (the kernel of evaluation at P), $F_1 F_1 \subseteq F_2$, and

$$F_1/F_2 \cong H^0(D', \mathcal{O}_{D'}(-D')) = H^0(D, \mathcal{O}(2, 2)), \quad \dim = 9.$$

Since $\hat{\mathfrak{m}}^2 \subseteq F_2$, the cotangent space $\hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2$ surjects onto F_1/F_2 , whence $\dim T_P \bar{Y} \geq 9$. This inequality is all that the $\mathbb{P}^7$ obstruction needs.

Step 2: equality, and the published theorem that identifies the germ. The same H^1 -vanishing gives $F_n/F_{n+1} \cong H^0(D, \mathcal{O}(2n, 2n))$ for all n , with the multiplication of sections, so the graded ring $\bigoplus_n F_n/F_{n+1}$ is the anticanonical ring $A = \bigoplus_n H^0(D, \mathcal{O}(2n, 2n))$ of D . It is generated in degree one (monomials of bidegree $(2n, 2n)$ are products of monomials of bidegree $(2, 2)$; logs/c02 records the Hilbert function $(2n+1)^2$). Hence $F_n = \hat{\mathfrak{m}}^n + F_{n+1}$ for all n , and since the F_n define the $\hat{\mathfrak{m}}$ -adic topology (Artin–Rees, as in the proof of the theorem on formal functions), $F_2 = \hat{\mathfrak{m}}^2$: $\dim T_P = 9$ and $\text{mult}_P \bar{Y} = \deg D_8 = 8$. The cleanest published route is Gross, *loc. cit.*, Proposition 5.4: *an isolated rational Gorenstein threefold point of multiplicity $k \geq 5$ whose blow-up is smooth with smooth exceptional divisor E is analytically the cone over E* ; its proof uses Grauert’s linearisation theorem and $I_E/I_E^2 = \omega_E^{-1}$, so “cone over E ” means the anticanonical cone. The hypotheses hold for $(\bar{Y}, P)$: it is Gorenstein with $K_{\bar{Y}} = 0$ (a Calabi–Yau threefold in Kapustka’s sense), hence rational, and by Reid’s theorem quoted in the proof of Gross’s Theorem 5.2 a primitive type II contraction is the blow-up of the

singular point, here with $\tilde{X} = X$ and $E = D' \cong \mathbb{P}^1 \times \mathbb{P}^1$ of degree 8. This is the theorem that connects the local calculation to Kapustka's contraction.

Step 3: the cone is the quotient of a node. Write $D = \{q = 0\} \subset \mathbb{P}^3 = \mathbb{P}(U)$ with $q = xy - zw$; then $-K_D = \mathcal{O}_{\mathbb{P}^3}(2)|_D$, so $A_n = H^0(D, \mathcal{O}(2n))$ is the even part of $k[x, y, z, w]/(q)$, that is, the ring of invariants of ± 1 acting on the node $\{q = 0\} \subset \mathbb{A}^4$. Exactly over $\mathbb{Q}$ (logs/c02): the kernel of $k[a_{ij}] \rightarrow k[x, y, z, w]/(q)$, $a_{ij} \mapsto$ (product of the i th and j th coordinates), is the one linear form $\lambda = a_{01} - a_{23}$ together with twenty quadrics, and modulo λ it coincides with the ideal of D_8 after the parametrisation $(x, y, z, w) = (su, tv, sv, tu)$. The vertex has tangent dimension $10 - 1 = 9$; every other point of the cone has tangent dimension 3. $\square$

3 Consequences, and why they do not survive smoothing

Proposition 3.1. $\bar{Y}$ has no closed embedding into $\mathbb{P}^7$, or into any smooth 7-fold. In particular T is not very ample. The morphism $\varphi_{|T|} : \bar{Y} \rightarrow \mathbb{P}^7$ defined by the eight sections has differential of rank ≤ 7 at P , and its image Y is not normal at $\varphi_{|T|}(P)$: the normalisation $\nu : \bar{Y} \rightarrow Y$ satisfies $\dim_k \nu_* \mathcal{O}_{\bar{Y}} / \mathcal{O}_Y = 2$, so

$$\chi(\mathcal{O}_Y(n)) = P(n) - 2 = \frac{10}{3}n^3 + \frac{14}{3}n - 2.$$

Hence $[Y]$ and $[\text{SR}(\mathcal{M})]$ lie in different Hilbert schemes of $\mathbb{P}^7$.

Proof. The first sentence is $\dim T_P \bar{Y} = 9 > 7$ together with the surjection of cotangent spaces recalled above. “Very ample” means that the complete linear system $|T|$, here of dimension 7, embeds $\bar{Y}$; so T is not very ample. Eight sections give a morphism to $\mathbb{P}^7$ because $|T|$ is base point free: the seven sections $s x_i$ vanish exactly on D' (the x_i have no common zero on X') and w does not vanish on D' . At P , in the chart $w \neq 0$, the seven functions $x_i s / w$ lie in $\hat{\mathfrak{m}}$ and their images in $F_1 / F_2 = H^0(D, \mathcal{O}(2, 2))$ are the seven linear forms of $\mathbb{P}^6$ restricted to S , spanning the 7-dimensional V . So the differential misses a 2-dimensional part of T_P : this is precisely why eight sections do not embed. For the non-normality, $\hat{\mathcal{O}}_Y$ is the closed subalgebra of $\hat{\mathcal{O}}$ generated by those seven functions, and with the filtration induced from $F_\bullet$ its graded ring is $k \oplus V \oplus A_2 \oplus A_3 \oplus \dots$: in degree one only V , in degree $n \geq 2$ all of A_n because $k[S]_n = A_n$, that is, $h^1(I_S(n)) = 0$ for $n \geq 2$ (logs/c03). Hence $\dim \hat{\mathcal{O}} / \hat{\mathcal{O}}_Y = \dim A_1 / V = 2$. Finally $\chi(\mathcal{O}_{\bar{Y}}(nT))$ is a cubic polynomial with leading coefficient $T^3/6 = 20/6$, constant term $\chi(\mathcal{O}_{\bar{Y}}) = 0$, odd in n by Serre duality ($\omega = \mathcal{O}$), and value 8 at $n = 1$ (vanishing of higher cohomology of T on $\bar{Y}$, which has rational singularities); so it equals $P(n)$, and $\chi(\mathcal{O}_Y(n)) = P(n) - 2$ because $\nu_* \mathcal{O}_{\bar{Y}} / \mathcal{O}_Y$ is a skyscraper of length 2. The 2026-09-11 run's finite-field model of Y (logs/c04, over $\mathbb{F}_{32003}$) confirms $\dim T_P Y = 7$ and the Hilbert polynomial $P(n) - 2$. $\square$

The same fact seen globally: the section ring $R = \bigoplus_n H^0(\bar{Y}, nT)$ has the Hilbert function $1, 8, 36, 104, \dots$ of $k[\mathcal{M}]$, but needs two generators in degree 2 besides the eight in degree 1, because $\text{Sym}^2 H^0(T) \rightarrow H^0(2T)$, a map of 36-dimensional spaces, kills the two quadrics of X' . Locally, two of the nine cotangent directions at P are invisible to the eight sections. Both are the two units of linear normality lost when D_8 was projected from a line. R is a Gorenstein ring (vanishing plus $K_{\bar{Y}} = 0$), so $\bar{Y}$ is “arithmetically Gorenstein” in the weighted space $\text{Proj } R$, but not in $\mathbb{P}^7$, where its image is not even normal.

Why none of this persists after smoothing

The function $t \mapsto \max_{y \in Y_t} \dim T_y Y_t$ is upper semicontinuous: tangent dimensions can only drop when one moves away from the special fibre, and on a smooth fibre they equal 3 everywhere.

Example 3.2. The surface node $\{xy = z^2\} \subset \mathbb{A}^3$ has $\dim T_0 = 3$; the fibres $\{xy - z^2 = t\}$, $t \neq 0$, are smooth with $\dim T = 2$ at every point. Closer to home, Gross (*loc. cit.* 5.5) smooths the cone of Theorem 2.1 by hyperplane sections of the cone over the second Veronese $v_2(\mathbb{P}^3) \subset \mathbb{P}^9$. Let $W_t \subset \mathbb{A}^{10}$ be the set of symmetric 4×4 matrices A of rank ≤ 1 with $\lambda(A) = t$, where $\lambda(A) = a_{01} - a_{23}$. Then $W_0 = (\bar{Y}, P)$, and for $t \neq 0$, $W_t = \{xy - zw = t\} / \{\pm 1\}$ is smooth because ± 1 acts freely on the smooth affine quadric. Over $\mathbb{Q}$ (logs/c02): $\dim T = 9$ at the vertex of W_0 and $\dim T = 3$ at two sample points of W_1 . Embedding dimension 9 becomes 3 the moment $t \neq 0$.

The property that matters for Y_t is therefore not local. It is whether the *eight* sections of T_t embed Y_t with projectively normal image, that is, whether

$$(\star) \quad \mu_t : \text{Sym}^2 H^0(Y_t, T_t) \longrightarrow H^0(Y_t, 2T_t) \quad (36 \rightarrow 36)$$

is an isomorphism, equivalently whether Y_t is aG in $\mathbb{P}^7$. Very ampleness and projective normality are open conditions in the family: they fail at $t = 0$ (Proposition 3.1), and openness gives no information in the other direction. Semicontinuity gives $\dim \ker \mu_t \leq \dim \ker \mu_0 = 2$, which is the wrong inequality. The singular model is silent on $(\star)$.

4 The limited relevance to $\text{SR}(\mathcal{M})$

Proposition 4.1. *Let $\mathcal{X} \rightarrow (B, 0)$ be a flat family of closed subschemes of $\mathbb{P}^7$ with $\mathcal{X}_0 = \text{SR}(\mathcal{M})$ (an embedded deformation). Then every fibre $\mathcal{X}_b$, b near 0, is aG with h -vector $(1, 4, 10, 4, 1)$, lies on no quadric, is cut out by 16 cubics, and has $\omega_{\mathcal{X}_b} \cong \mathcal{O}_{\mathcal{X}_b}$. If it is smooth and connected it is a Calabi–Yau threefold of degree 20 with $h^0(\mathcal{O}(1)) = 8$, $H^3 = 20$, $c_2 \cdot H = 56$.*

Proof. Flatness fixes the Hilbert polynomial P . Write $H_b(k)$ for the Hilbert function of $S/I_{\mathcal{X}_b}$. Semicontinuity gives $\dim(I_{\mathcal{X}_b})_k \leq \dim(I_{\mathcal{M}})_k$ and $h^0(\mathcal{O}_{\mathcal{X}_b}(k)) \leq h^0(\mathcal{O}_{\mathcal{X}_0}(k)) = H_0(k)$ (the last equality because $k[\mathcal{M}]$ is Cohen–Macaulay), so $H_0(k) \leq H_b(k) \leq h^0(\mathcal{O}_{\mathcal{X}_b}(k)) \leq H_0(k)$: the Hilbert function is constant and the fibres are projectively normal. With constant Hilbert function the graded Betti numbers are upper semicontinuous, $\beta_{i,j}(\mathcal{X}_b) \leq \beta_{i,j}(\text{SR}(\mathcal{M}))$, so the resolution of $\mathcal{X}_b$ has length $\leq 4 = \text{codim}$ (hence exactly 4, Cohen–Macaulay) and last Betti number ≤ 1 (hence 1: Gorenstein), sitting in degree 8. This gives the h -vector, the absence of quadrics, the 16 cubics, and $\omega \cong \mathcal{O}$. The invariants follow from the Hilbert polynomial by Riemann–Roch on a smooth fibre. $\square$

The 2026-09-11 run’s `REPORT.md`, Theorem 5.1, asserts more: that every *abstract* smoothing of $\text{SR}(\mathcal{M})$ is of this embedded form, because $h^1(\mathcal{O}) = h^2(\mathcal{O}) = 0$ (Hochster’s formula, $\mathcal{M}$ a homology sphere) lets $\mathcal{O}(1)$ extend uniquely to the family. I have not re-verified that step; it is not needed for the smoothing under review, which is constructed as a deformation of the ideal $I_{\mathcal{M}}$ inside $\mathbb{P}^7$.

What Theorem 2.1 and Proposition 3.1 actually give for the comparison:

- (i) The two *singular* models are incomparable. $\bar{Y}$ is irreducible and normal with one point of embedding dimension 9 and is not a subscheme of $\mathbb{P}^7$; its $\mathbb{P}^7$ image Y has Hilbert polynomial $P(n) - 2$. $\text{SR}(\mathcal{M})$ is a reducible aG subscheme of $\mathbb{P}^7$ with Hilbert polynomial $P(n)$. No flat family inside $\mathbb{P}^7$ connects Y or $\bar{Y}$ to $\text{SR}(\mathcal{M})$, so no degeneration of Kapustka’s construction to $\text{SR}(\mathcal{M})$ can be read off from the objects Kapustka writes down. (This is a structural fact, not a failed search.)
- (ii) Any comparison must go through the smooth fibres, and there it is exactly $(\star)$. *Conditionally:* if $(\star)$ holds for general t , then $Y_t \subset \mathbb{P}^7$ is an aG Calabi–Yau threefold of degree 20 with h -vector $(1, 4, 10, 4, 1)$ and $h^{1,1} = 1$, hence a point of the same Hilbert scheme as any embedded smoothing of $\text{SR}(\mathcal{M})$, and the question becomes whether the two lie on a common component. If $(\star)$ fails, no embedded smoothing of $\text{SR}(\mathcal{M})$ has fibres isomorphic to Kapustka’s Y_t with $T_t = \mathcal{O}(1)$, by Proposition 4.1. Neither alternative is decided here, and nothing here says that Kapustka’s Y_t is non-aG or abstractly deformation-inequivalent to a fibre of an $\text{SR}(\mathcal{M})$ smoothing.
- (iii) The one published aG degree-20 family in $\mathbb{P}^7$ (the determinantal Gulliksen–Negård threefold, HP “GN variety”) has $h^{1,1} = 2$, so it is a different family from Kapustka’s $h^{1,1} = 1$ one, whatever the answer to $(\star)$.

Conclusion and next question

Conclusion. Kapustka’s normal contraction $\bar{Y}$ has a single singular point P , analytically the cone over the anticanonical $\mathbb{P}^1 \times \mathbb{P}^1$, i.e. $\{xy = zw\}/\{\pm 1\}$, with $\dim T_P \bar{Y} = 9$ (the lower bound ≥ 9 is elementary from the theorem on formal functions and $H^1(\mathcal{O}_D(2, 2)) = 0$; equality and the analytic model are Gross’s Proposition 5.4). Hence $\bar{Y}$ embeds in no smooth 7-fold, T is not very ample, the eight sections map $\bar{Y}$ onto a non-normal $Y \subset \mathbb{P}^7$ with Hilbert polynomial $P(n) - 2$, and the singular models of the two constructions cannot be compared inside $\mathbb{P}^7$. This is a statement about the special fibre only; embedding dimension drops to 3 on every smooth fibre, and whether the eight sections of T_t embed Y_t as an aG threefold in $\mathbb{P}^7$ (property $(\star)$) is neither implied nor excluded by it.

Next question, for Kapustka. In Theorem 3.4 the degree-15 fibres are embedded by $|2T_t|$. For row 8, is $\mu_t : \text{Sym}^2 H^0(T_t) \rightarrow H^0(2T_t)$ an isomorphism for general t , i.e. does $|T_t|$ embed Y_t in $\mathbb{P}^7$ with projectively normal image? Equivalently: does the general member of the family lie on no quadric of $\mathbb{P}^7$? The 2026-09-11 run claims a first-order obstruction inside the weighted model (not re-verified here) and reduces the decision to a second-order term.